

BIOTECHNOLOGY & GENETIC MODIFICATION

biotechnology & genetic modification

- * bacteria are useful in biotechnology and genetic modification due to their rapid reproduction rate and their ability to make complex molecules

Why are they useful?

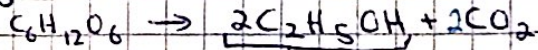
- very few ethical concerns over their manipulation and growth
- the presence of plasmids (ideal way of transferring DNA)

• Yeast:

- a single-celled fungus that uses sugar as its food source
- when it respire ethanol and CO_2 is produced (fermentation)
- respire anaerobically (with no oxygen)

→ ethanol for biofuel

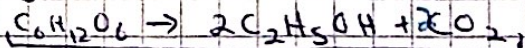
glucose → ethanol + carbon dioxide



- used as a biofuel

→ bread-making

glucose → ethanol + carbon dioxide



- the dough rises due to CO_2 which gets trapped as the yeast eats the sugar
- temperature is important as respiration is controlled by enzymes

• Fruit juice production:

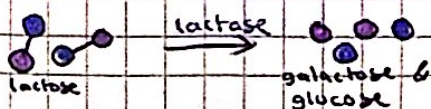
- pectinase is an enzyme that breaks down pectin which is found in plant cell walls, to extract more fruit juice easily.

• Biological washing powders:

- used to break down organic substances, mainly protease and lipase. Due to the need of being used with hot water, enzymes must have high optimum temps. ($\sim 70^\circ\text{C}$)
- ↳ extracted from thermophilic bacteria.

• Lactose-free milk:

- milk can be made lactose free by adding the enzyme lactase to it and leaving it for a while to allow the enzyme to break down the lactose



• Insulin:

- insertion of human cells with healthy insulin genes to bacteria
- ↳ chromosomes of human cells are removed
- ↳ insulin gene from chromosome removed via restriction enzyme
- ↳ plasmids removed from bacterial cell
- ↳ plasmids cut open by the same enzyme
- ↳ human insulin gene inserted into plasmids using ligase enzymes
- ↳ plasmids returned to bacterial cell
- ↳ bacterial cell reproduces in fermenter making more insulin

Penicillin:

- antibiotic made by a fungus called penicillium
- sugar, ammonium salt + penicillium added to fermenter
 - ↳ sugar allows penicillium to respire
 - ↳ ammonium allows penicillium to make protein + nucleic acid

Mycoprotein:

- protein-rich food suitable for vegetarians made by *Fusarium* (fungus)
 - ↳ made in fermenters
 - ↳ grows in aerobic conditions + presence of glucose (respire)
 - ↳ fungus grows and multiplies in fermenter

Fermenter:

- probes to monitor temperature and pH
- Oxygen for respiration
- Stirrer keeps bacteria suspended.

Genetic modification:

changing the genetic material of an organism by removing, changing or inserting individual genes

Process of genetic modification:

- isolate the DNA making up a human gene using restriction enzyme that form sticky ends
- cut the bacterial plasmid DNA with the same restriction enzymes, forming complementary sticky ends
- insert the human DNA into bacterial plasmid DNA using DNA ligase to form a recombinant plasmid
- insert the recombinant plasmids into bacteria
- the bacteria containing the recombinant plasmids will multiply
- the bacteria will express the human gene to make the human protein

Examples of genetic modification:

- insertion of human genes into bacteria to produce human proteins
- insertion of genes into crop plants to confer resistance to insect pests
- insertion of genes into crop plants to confer resistance to herbicides
- insertion of genes into crop plants to improve nutritional qualities

Advantages

- ① increase crop yield
- ② reduced cost for food/drugs
- ③ reduced needs for pesticides

Disadvantages

- ① increase in toxin level of soil
- ② social and ethical concerns
- ③ allergic reactions